

# Anil Reddy Mokalla

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## **Summary:**

Expert engineer skilled in designing comprehensive validation frameworks, debugging complex system-level issues, and automating cross-platform processes using Python and Shell scripting. Highly proficient in managing version control via Git and Bitbucket and optimizing workflows within Agile environments. Specialized in the semiconductor domain, focusing on SoC validation for high-performance SSDs and NVMe drive protocol compliance.

## **Skills:**

Robotic Framework, PyCharm, Visual Studio, Rundeck, Git Hub, Jira, Jenkins, Python, C, Quality Assurance, Json, Linux, Windows, kernel, system debugging, Manual Testing, Documentation skills, Log Analysis, Automation Testing, Test Planning, systems Testing, Thermal chamber testing, Bitbucket, NVME spec verification, UART, I2c, FPGA, lspcie, nvme-cli, FTL, SMART, PLP, Teledyne LeCroy, VIAVI.

## **Experience: Senior Software Engineer:**

**Mirafr Technologies**

Hyderabad, India

March 2026 - present

## **Experience: Software QA Engineer:**

**Airaa Corporates LLP - Micron as a client**

Hyderabad, India

Nov 2021 - Feb 2026

### **Project: ss/nvme/firmware validation:**

- Designed and distributed comprehensive test cases to cover workloads, ensuring timely completion of the checkout process and providing extensive test coverage for vendor-specific test cases across various OEMs.
- Configured and installed Linux operating systems, incorporating required drivers and building site packages. Conducted thorough debugging of system-level issues to maintain optimal system performance and stability. Registered DUTs in the Long run IO framework for test execution using the Robot framework.
- Tracked and managed issues using Jira, ensuring timely follow-ups and updates. Performed error injection and log collection for debugging, demonstrating a proactive approach to issue resolution.
- Developed Python scripts for file handling and utilized Shell scripts to reduce time consumption in testing. Conducted basic debugging for NVMe driver and system-level issues, analyzing UART logs using Tera Term and CrystalDisk.
- Created comprehensive Confluence pages detailing project overview, requirements, design, testing, and issue tracking. Maintained and updated test cases using GitHub, ensuring relevance and accuracy.
- Created and managed pull requests, ensuring effective collaboration and quality code merging processes. Made code changes and tracked updates efficiently using Bitbucket repositories

- Reworked and optimized PCIe adaptor integration: Evaluated and refined PCIe adaptor designs to improve signal integrity and throughput, addressing compatibility challenges and ensuring seamless communication between NVMe SSDs and host controllers.
- Skilled in identifying and debugging temperature-induced hardware failures by correlating thermal chamber logs with SoC performance metrics and NVMe protocol traces, chamber testing. Power management in different power states with workloads. Smmps, DMM, Thermal Threshold checking

## **Technical Achievements:**

- Scripting Efficiency: Developed a Python-based Log Parser that extracts critical error signatures from large dump logs, reducing manual debugging time from hours to minutes.
- FW Update Automation: Created a JSON-driven automation script and scheduler that streamlined the firmware update process across 50+ concurrent systems, saving 15+ man-hours per week.
- Test Cycle Optimization: Reduced end-to-end execution time by 30% through advanced Shell scripting and task parallelization in Linux.

## **Project:**

- Streamlined SSD performance and reliability verification by developing a Python-based automated testing framework, increasing test accuracy and throughput.
- Executed end-to-end validation for 4 NVMe SSD procurement cycles, ensuring strict adherence to NVMe 1.4/2.0 specifications.
- Identified and reported critical defects in firmware bug, controller registers, and Power Loss Protection (PLP) hardware using NFI and GBB reporting, preventing sub-standard hardware deployment.

## **Certification courses:**

Embedded systems – (vector India, Hyderabad)

Certified in Embedded Systems Programming with hands-on experience in designing, developing, and testing firmware for microcontrollers and processors. Demonstrated expertise in real-time operating systems (RTOS), peripheral interfaces, and debugging tools to ensure system reliability and performance.

**Education: B Tech in Electrical and Electronics Engineering                      2017- 2021**  
Visvesvaraya College of Engineering and Technology Affiliated by JNTUH

**MTech in Embedded systems**

**2023-2025      BITS Pilani WILP**

